

Project 2: Sales Forecasting & Demand Prediction (Time Series + Dashboard)

Objective

Build an end-to-end time series forecasting solution to predict future retail sales using historical data, evaluate the model using standard metrics, and present business insights through an interactive dashboard.

Dataset Overview

A synthetic retail dataset was created to simulate real-world sales behavior across multiple cities and product categories. It contains daily transactional records with seasonality (Oct–Dec peak demand), promotion effects, and weekday/weekend behavior.

Column	Description
Date	Transaction date (daily records)
Store Location	City where the store is located
Product Category	Product segment (Electronics, Clothing, Groceries, Furniture, Cosmetics)
Units Sold	Quantity sold per day per store & category
Sales Amount	Total revenue (Units Sold × price)
Promotions/Discounts	Promotion flag (0 = No promo, 1 = Promo)
Day Type	Weekday or Weekend

Methodology

- Data Preprocessing:** Converted date column to datetime format, checked missing values, and aggregated daily sales into monthly sales totals for time-series forecasting.
- Exploratory Data Analysis (EDA):** Analyzed monthly trend, sales by category/location, and promotion impact on sales.
- Forecasting Model:** Prophet forecasting was used to capture trend and seasonality automatically.
- Evaluation:** Performed time-aware train-test split with the last 6 months as test set. Metrics used: MAE, RMSE, MAPE.

Model & Evaluation Metrics

The Prophet model was trained on monthly aggregated sales. Model performance was measured using:

- **MAE** (Mean Absolute Error)
- **RMSE** (Root Mean Squared Error)
- **MAPE** (Mean Absolute Percentage Error)

Note: Final metric values are automatically computed in the provided Colab notebook.

Key Business Insights

- Sales show seasonal peaks during Oct–Dec (festival/holiday demand), requiring higher stock levels in those months.
- Groceries and Clothing contribute high and consistent revenue, making them priority categories for inventory planning.
- Promotions significantly increase average units sold and revenue, indicating strong responsiveness to discounts.

- Some store locations consistently outperform others—marketing budget and stock allocation should be optimized city-wise.
- Forecasted sales for the next 6 months support proactive decisions in staffing, logistics, procurement, and campaign planning.

Dashboard Recommendation (Power BI / Tableau)

Page 1: Sales Overview — KPIs, monthly trend, sales by category & location.

Page 2: Promotions & Demand — promo vs non-promo impact, units trend, filters for city/category.

Page 3: Forecast — actual vs forecast line chart, next 6 months forecast table with confidence bounds.

The file **monthly_sales_forecast_output.csv** generated by the notebook can be imported into the dashboard for forecasting visuals.

Conclusion

This project demonstrates an end-to-end retail forecasting workflow—from data preparation and analysis to model forecasting and dashboard-based decision support. The final solution helps businesses anticipate demand, reduce inventory risk, and plan promotions effectively.